

Notes: Lyonnet and Chrétien (RFS, 2026)

Why Do Traditional and Shadow Banks Coexist?

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1 Big picture

- **Question.** Why do *traditional banks* and *shadow banks* coexist even though they provide similar money-like liabilities, and how does their interaction matter in crises and under regulation?
- **Setting.** The paper combines a three-date equilibrium model of traditional banks (T-banks) and shadow banks (S-banks) with evidence from the 2007 financial crisis and the 2020 COVID-19 stress episode.
- **Core challenge.** Shadow banks avoid regulation, which seems to give them a cost advantage, but they cannot rely on deposit insurance. Traditional banks can issue insured claims, but they pay a regulatory cost. A convincing coexistence theory has to explain why *both* survive and why flows run from S-banks to T-banks in crises.
- **Main theoretical contribution.** The paper introduces asset trade between T-banks and S-banks in crises. This possibility creates a *complementarity*: shadow banks can issue riskless short-term debt in good times because they expect traditional banks to buy their risky assets in bad times, while traditional banks value the option to buy those assets at a discount in crises.
- **Main equilibrium result.** T-banks and S-banks coexist when crisis asset trade is feasible and an equilibrium crisis price can simultaneously (i) compensate T-banks for regulation and (ii) support S-banks' debt capacity enough to make expected profits equal across bank types.
- **Main empirical facts.** In both 2007 and 2020, shadow-bank liabilities contracted, large traditional banks experienced major deposit inflows, shadow-bank assets moved onto traditional-bank balance sheets, and the transferred assets traded under fire-sale pressure.
- **Main regulation result.** Increasing T-banks' debt capacity can *increase* the relative size of the shadow-banking sector. The reason is a new **complementarity effect**: stronger T-banks bid up S-bank asset prices in crises, which relaxes S-banks' ex ante debt capacity. This can dominate the usual **regulatory-arbitrage effect** that pushes activity toward T-banks.
- **Main welfare result.** In the decentralized equilibrium, there are too many S-banks and too few T-banks relative to the social planner because bankers do not internalize how their entry decisions affect the crisis asset price.
- **Why this paper matters.** The paper gives a parsimonious equilibrium foundation for why shadow banks do not disappear when deposit insurance exists, why crises involve flows from shadow to traditional intermediation, and why tightening or relaxing traditional-bank regulation can have unintended consequences for the size of shadow banking.

2 Data and measurement (key objects)

2.1 Observed aggregate banking-sector evidence

- **Traditional-bank sector.** The paper measures T-bank size mainly using *private depository institutions* from the Financial Accounts of the United States (FAUS, table L.110).
- **Shadow-bank sector.** S-bank liabilities are measured as the sum of money market mutual fund shares, repurchase agreements not involving commercial banks or the Fed, and financial-sector

commercial paper. This is meant to capture institutions that back money-like claims with risky assets.

- **Shadow-bank runs.** To illustrate runs in 2007 and 2020, the paper uses Investment Company Institute data on *prime* and *government* money market mutual funds.
- **Deposits and MBS holdings of large T-banks.** Weekly Federal Reserve H.8 releases are used to track deposit inflows and mortgage-backed securities (MBS) purchases by the largest U.S. traditional banks.
- **Asset-transfer evidence.** Holdings of agency- and GSE-backed securities by T-banks and S-banks are built from the FAUS. Shadow-bank holdings are measured as total holdings excluding households, the government, traditional banks, and the Federal Reserve.
- **Fire-sale evidence.** Bloomberg data are used to track agency-MBS spreads and the stock prices of mortgage REITs during the COVID-19 turmoil.

2.2 Bank-level regression data for the crisis asset-transfer test

- **Unit of observation.** Bank holding company.
- **Sample period.** The main regression evidence focuses on the global financial crisis, comparing changes from 2007q4 to 2009q1.
- **Dependent variable.** MBS purchases by traditional banks over the crisis period, normalized by total assets as of 2007q4.
- **Data source.** Quarterly Call Reports plus calibration assumptions from He, Khang, and Krishnamurthy-style measurement of actual MBS purchases.
- **Measurement scenarios.** The paper reports three variants of MBS-purchase measurement: two adjusted scenarios and one “naive” change-in-holdings measure, with and without accounting for ABCP conduit consolidation.
- **Sample size.** Table 1 reports 3,954 observations in each specification.

2.3 Unobservables and timing in the model

- **Dates.** There are three dates, $t = 0, 1, 2$.
- **States.** At $t = 1$, either a good state G arrives with probability p or a crisis state B arrives with probability $1 - p$. At $t = 2$, if state G occurs then payoff state GG is known; if crisis state B occurs, then payoff state BG arrives with probability q and BB with probability $1 - q$.
- **Endogenous crisis asset price.** The key equilibrium object is the crisis asset price $p_{1,B}$, which determines how much S-banks can borrow ex ante and whether T-banks are willing to keep slack ex ante so that they can buy assets in the crisis.
- **Bank-type composition.** The endogenous share of bankers choosing a shadow bank is denoted χ^S .

2.4 Empirical applications of the model

- The evidence section documents three crisis facts: liabilities leaving S-banks, assets moving from S-banks to T-banks, and fire-sale price pressure.
- The model is then used for comparative statics on: T-bank debt capacity, regulatory costs, crisis probability, inelastic demand for short-term debt, liquidity regulation, and the planner's preferred mix of T- and S-banks.

3 Models and empirical specifications (all equations + interpretation)

3.1 Households, bankers, and preferences

The economy has one good, two groups of agents, and three dates. A unit mass of bankers starts with net worth n and can choose to set up either a traditional bank or a shadow bank. Households are endowed with goods at $t = 0, 1$, cannot invest directly in risky assets, and buy only claims issued by banks. Their utility is

$$U_0 = C_0 + \mathbb{E}_0[U_{1,\omega_1} + U_{2,\omega_2}], \quad U_{1,\omega_1} = C_{1,\omega_1} + \mathbb{E}_{1,\omega_1}[U_{2,\omega_2}], \quad U_{2,\omega_2} = C_{2,\omega_2}. \quad (1)$$

- **Interpretation.** Households are effectively risk neutral and value short-term safe claims, which is what gives banks a role as issuers of money-like liabilities.
- **Bankers' choice.** Each banker chooses whether to start a T-bank or an S-bank and invests all net worth in that bank in equilibrium.

3.2 Risky assets and the timeline of the model

At $t = 0$, banks invest in risky assets. At $t = 1$, the economy is either in a normal state G or in a crisis state B . The asset payoff structure is:

- if state G occurs at $t = 1$, then the $t = 2$ state is GG for sure and assets pay $R > 0$;
- if crisis state B occurs at $t = 1$, then at $t = 2$ assets pay $r > 0$ in state BG with probability q , and 0 in state BB with probability $1 - q$.

Banks choose assets and debt at $t = 0$, and in state B at $t = 1$ they can liquidate assets, buy other banks' assets, and issue short-term debt subject to their funding technology.

- **Interpretation.** The state B is the crisis in which asset trade matters. T-banks and S-banks have the same risky assets, but different ways of financing them.
- **Key friction.** Households want *riskless short-term debt*, while assets are risky. This makes balance-sheet structure central.

3.3 Differences between traditional and shadow banks

The paper's main institutional assumption is:

Assumption 1 (Differences between traditional and shadow banks).

- T-banks can issue debt backed by deposit insurance up to face value k per bank.
- T-banks incur a regulatory cost and receive only $\delta \in [0, 1]$ of asset payoffs at $t = 2$.
- **Interpretation of k .** The parameter k is the reduced-form debt capacity of T-banks in crises. It can represent deposit insurance directly, or more broadly the ability of T-banks to issue safe debt using government support or central-bank access.
- **Interpretation of δ .** T-banks' regulation acts like a haircut on asset returns. Shadow banks avoid this cost but cannot insure their liabilities.
- **Core trade-off.** T-banks have safer funding but lower net asset return. S-banks have cheaper intermediation but fragile funding.

3.4 Parametric restrictions that generate coexistence

The paper imposes several economically meaningful restrictions:

$$\delta(pR + (1 - p)qr) > 1, \quad (2)$$

$$qr < 1, \quad (3)$$

$$\epsilon > 1 - \delta. \quad (5)$$

It also requires:

$$\delta < \frac{p}{\epsilon(p(R - 1) + (1 - p)qr) + p}, \quad (6)$$

and a debt-capacity interval

$$k \in [\underline{k}, \bar{k}]. \quad (7)$$

- (2) says the risky asset is valuable ex ante even for regulated T-banks.
- (3) says a crisis is truly a bad-news state: expected returns fall in state B .
- (5) says liquidation is sufficiently costly that in the good state it is not optimal for T-banks to sell assets to S-banks.
- (6) ensures gains from trade in the crisis asset market: the regulatory cost is high enough that T-banks want to keep slack to buy S-bank assets in a crisis.
- (7) ensures T- and S-bank expected profits can be equalized. If k is too low, T-banks are too weak; if k is too high, S-banks cannot compete.

3.5 Shadow-bank problem and Proposition 1

In the crisis state, shadow banks can only survive by liquidating assets. Their crisis payoff is

$$\pi_{1,B}^S(I_0^S, D_0^S, p_{1,B}) = \max_{\alpha_{1,B}^S} \{ \alpha_{1,B}^S qr I_0^S + (1 - \alpha_{1,B}^S)(1 - \epsilon)p_{1,B} I_0^S - D_0^S \}, \quad (8)$$

subject to repayment of date-0 debt by liquidating enough assets.

At $t = 0$, shadow banks choose assets and debt taking the future crisis price as given:

$$\pi_0^S(p_{1,B}, n^S) = \max_{D_0^S, I_0^S} \left[p\pi_{1,G}^S(I_0^S, D_0^S) + (1-p)\pi_{1,B}^S(I_0^S, D_0^S, p_{1,B}) + (D_0^S + n^S - I_0^S) \right] \quad (1)$$

subject to

$$I_0^S \leq D_0^S + n^S, \quad D_0^S \leq (1-\epsilon)p_{1,B}I_0^S, \quad D_0^S \leq RI_0^S.$$

Proposition 1.

- For low future crisis prices $0 \leq p_{1,B} < p_1^S$, S-banks do not issue riskless debt at $t = 0$ and invest only their equity.
- For high future crisis prices $p_1^S < p_{1,B} < qr/(1-\epsilon)$, S-banks issue as much riskless debt as they can, backed by the liquidation value of assets in a crisis.
- **Interpretation.** S-banks' ex ante debt capacity is entirely pinned down by what they can raise by selling assets in state B . They are fragile by construction.
- **Economic meaning.** S-banks can mimic money creation only because they expect somebody else—namely T-banks—to absorb their assets in bad times.

3.6 Traditional-bank problem and Proposition 2

In the good state, T-banks simply repay creditors by rolling debt. In the crisis state they can buy fire-sold assets, financing those purchases with debt backed by deposit insurance:

$$\pi_{1,B}^T(I_0^T, D_0^T, p_{1,B}) = \max_{\alpha_{1,B}^T, D_{1,B}^T, I_{1,B}^T} \left[(\delta qr - p_{1,B})I_{1,B}^T + \alpha_{1,B}^T \delta qr I_0^T + (1 - \alpha_{1,B}^T)(1 - \epsilon)p_{1,B}I_0^T - D_0^T \right] \quad (9)$$

subject to

$$D_{1,B}^T \leq k, \quad D_{1,B}^T \leq \delta qr(\alpha_{1,B}^T I_0^T + I_{1,B}^T).$$

At $t = 0$, T-banks solve

$$\pi_0^T(p_{1,B}, n^T) = \max_{D_0^T, I_0^T} \left[p\pi_{1,G}^T(I_0^T, D_0^T) + (1-p)\pi_{1,B}^T(D_0^T, I_0^T, p_{1,B}) + (D_0^T + n^T - I_0^T) \right].$$

Proposition 2.

- For low future crisis prices, T-banks do *not* issue debt at $t = 0$ and invest using equity only so that they preserve slack for crisis asset purchases.
- For high future crisis prices, T-banks issue as much debt as possible at $t = 0$ and roll it over in the crisis using deposit insurance.
- **Interpretation.** T-banks sometimes optimally keep balance-sheet capacity unused ex ante because buying S-bank assets in a crisis is sufficiently profitable.
- **Why this matters.** This is the source of complementarity. T-banks become valuable to S-banks precisely because they are potential buyers of fire-sold assets.

3.7 The crisis asset market and Proposition 3

Let $\chi^S \in [0, 1]$ denote the fraction of bankers who choose S-banks. In crisis state B , T-bank aggregate asset demand is

$$D(p_{1,B}) = \frac{k}{p_{1,B}}(1 - \chi^S) \quad \text{if } p_{1,B} < p_{1,L}^T, \quad (10)$$

and S-bank aggregate asset supply is

$$S(p_{1,B}) = \frac{n}{1 - (1 - \epsilon)p_{1,B}}\chi^S \quad \text{if } p_{1,B} > p_1^S. \quad (11)$$

Proposition 3 (Coexistence). A unique equilibrium exists in which

- bankers invest all endowment in one bank type at $t = 0$,
- no assets are traded in the good state,
- in the crisis state S-banks sell assets to T-banks,
- the crisis asset price makes bankers indifferent between starting a T-bank and starting an S-bank.
- **Interpretation.** The coexistence price is the one that both clears the crisis asset market and equalizes expected profits across bank types.
- **Complementarity.** The more bankers choose T-banks, the higher the crisis asset price and the easier it is for S-banks to issue debt ex ante. The more bankers choose S-banks, the more profitable crisis asset purchases become for T-banks.
- **No-deposit-insurance benchmark.** Without deposit insurance, this complementarity disappears and only one bank type survives.

3.8 Empirical specification for crisis asset transfers

To test the link between T-bank funding inflows and crisis asset purchases, the paper estimates regressions of the form

$$\text{MBSPurchases}_b = \beta_1 \Delta \text{InsuredDeposits}_b + \beta_2 \Delta \text{UninsuredDeposits}_b + \beta_3 \Delta \text{LiquidityRatio}_b + \beta_4 \Delta \text{Credit}_b + \beta_5 \Delta \text{CapitalRatio}_b \quad (2)$$

- **Interpretation.** The question is whether traditional banks that experienced larger insured-deposit inflows bought more MBS during the crisis.
- **Main coefficient.** Across specifications, the coefficient on insured-deposit growth is about 0.22–0.23 and highly significant, while the coefficient on uninsured-deposit growth is near zero in most specifications.
- **Economic meaning.** This is consistent with the model’s mechanism: T-banks funded crisis asset purchases with insured liabilities.

3.9 Comparative statics on regulation and crisis risk

Proposition 4 (Changes in T-bank debt capacity). In the coexistence equilibrium, an increase in T-banks' debt capacity k can increase the relative size of the shadow-banking sector χ^S .

- **Regulatory-arbitrage effect.** A higher k lets T-banks operate on a larger scale, which directly raises the incentive to become a T-bank.
- **Complementarity effect.** A higher k lets T-banks bid more aggressively for S-bank assets in crises, which raises the crisis asset price and therefore relaxes S-banks' ex ante debt capacity.
- **Baseline result.** In the paper's baseline model, the complementarity effect dominates, so increasing k shifts entry *toward* S-banks rather than toward T-banks.

The paper also shows:

- a higher regulatory cost for T-banks (lower δ) raises the equilibrium share of S-banks;
- a higher crisis probability (lower p) lowers the equilibrium share of S-banks because expected crisis fire-sale losses become more severe;
- if the crisis is too likely or too unlikely, only one bank type exists because the gains from trade in crises disappear.

3.10 Extensions: inelastic debt demand and liquidity regulation

Under an extension with inelastic demand for short-term debt, the equilibrium rate on bank debt becomes, in a simple symmetric case,

$$y_t^* = 1 - \frac{1}{D_t^T + D_t^S}. \quad (12)$$

Corollary 4.1. With inelastic demand for short-term debt, an increase in T-banks' debt capacity can instead *decrease* the relative size of S-banks when safe-claim scarcity is strong enough and regulatory arbitrage dominates complementarity.

- **Interpretation.** Once debt demand is inelastic, shrinking T-bank debt capacity can create scarcity of safe claims, lower funding rates, and strengthen the conventional story that tighter T-bank regulation pushes activity toward S-banks.
- **Empirical message.** Whether regulation grows or shrinks shadow banking is therefore not universal; it depends on market conditions.

The paper also studies liquidity regulation by requiring T-banks to hold liquid assets:

$$l \geq \rho I_0^T, \quad (15)$$

which changes the date-0 budget constraint to

$$I_0^T(1 + \rho) \leq D_0^T + n. \quad (16)$$

- **Interpretation.** Liquidity requirements both lower T-banks' effective return on risky assets and increase the liquidity they bring into the crisis state.
- **Result.** In the model these effects both push toward a larger S-bank sector.

3.11 Social planner and Proposition 5

The planner chooses the fraction of S-banks to maximize aggregate banker surplus:

$$\chi^{S*} = \arg \max_{\chi^S \in [0,1]} \{ \chi^S \pi_0^S(\cdot, p_{1,B}^*) + (1 - \chi^S) \pi_0^T(\cdot, p_{1,B}^*) \}. \quad (17)$$

Proposition 5. When T-banks and S-banks coexist, the decentralized equilibrium has too many S-banks and too few T-banks relative to the social optimum.

- **Interpretation.** Individual bankers do not internalize the pecuniary externality they impose on the crisis asset price.
- **Mechanism.** More S-banks depress the crisis resale value of assets, which hurts all S-banks; more T-banks raise that price and support the debt capacity of all S-banks. This externality is ignored privately.
- **Policy implication.** Transfers from S-banks to T-banks, such as a tax on S-banks used to subsidize T-banks, can raise welfare in the coexistence region.

4 Identification and instruments (what makes the estimates credible)

4.1 What this paper is and is not identifying

- This is primarily a **theory paper with empirical validation**, not a reduced-form causal paper with an exogenous treatment and a clean instrument.
- The empirical evidence is therefore not meant to *prove* the model in a strict causal sense. Instead, it is meant to show that the central mechanisms the model emphasizes are strongly consistent with observed crisis behavior in 2007 and 2020.
- The central object is not a causal treatment effect, but the equilibrium mechanism that can jointly explain coexistence in normal times and flows from S-banks to T-banks in crises.

4.2 Why the evidence is informative for the model

- **Fact 1: liability flows.** Runs on shadow-bank liabilities and simultaneous deposit inflows into traditional banks match the funding asymmetry in the model: only T-banks can scale riskless debt issuance in stress.
- **Fact 2: asset flows.** The model's distinctive prediction is not just liability substitution but *asset trade* from S-banks to T-banks in crises. The paper documents exactly this for both the 2007 and 2020 episodes.
- **Fact 3: fire-sale prices.** T-banks buy S-bank assets only if prices are sufficiently low. The observed price pressure in mortgage-linked assets is therefore essential because it lines up with the model's crisis-market-clearing logic.
- **Timing relative to the Fed.** The asset and liability flows begin before large-scale Fed MBS purchases in 2008, which helps the paper argue that the private-sector mechanism existed independently of the later public intervention.

4.3 Why Table 1 is the closest thing to a model-based identification test

- Table 1 does not identify a structural parameter, but it does test a specific implication of the model: traditional banks that received more insured-deposit inflows should buy more MBS in the crisis.
- The positive and significant insured-deposit coefficient, together with the weak role of uninsured deposits, is precisely the pattern the model predicts.
- The result is strongest when the dependent variable is adjusted for repayment, losses, and conduit consolidation, which is important because raw changes in holdings can be misleading during crises.

4.4 Model-based identification of coexistence

- Conditions (6) and (7) are the key model restrictions. They define the parameter region in which crisis asset trade is feasible and expected profits can be equalized across T- and S-banks.
- This is what makes coexistence a genuine equilibrium outcome rather than an assumed feature of the model.
- Figure 11 makes this especially clear: if regulation is too light or debt capacity is too large/small, the complementarity needed for coexistence breaks down and only one bank type survives.

4.5 Relation to the literature

- Relative to Hanson et al. (2015), the novelty is that the crisis discount is endogenized through equilibrium interaction between T-banks and S-banks rather than taken from an exogenous second-best asset market.
- Relative to LeRoy and Singhania (2020), the paper focuses on aggregate crisis risk and on assets selling at a *discount* to T-banks rather than at a premium under subsidized insurance.
- Relative to the standard regulation-and-shadow-banking literature, the paper's key departure is that T-banks and S-banks are *strategic complements*, not just substitutes. That is exactly why regulation can have unintended effects.

5 Tables and figures

5.1 Figure 1 and Figure 2: asset payoffs and model timeline

Read:

- Figure 1 defines the basic crisis technology: assets are valuable in normal times but risky in the crisis state, paying r with probability q and 0 otherwise.
- Figure 2 makes the sequential logic of the paper transparent: bank type choice and balance-sheet choice happen at $t = 0$, crisis trading and debt rollover happen at $t = 1$, and repayments occur at $t = 2$.

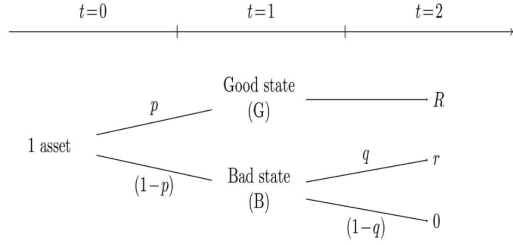


Figure 1
Asset payoff.

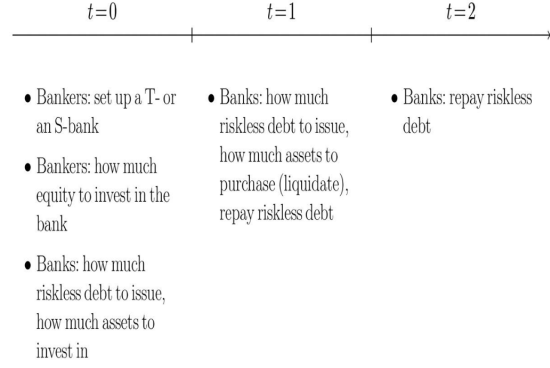


Figure 2
Timeline of the model.

- The entire coexistence result comes from the fact that crisis-period asset trade happens *after* initial balance-sheet choices have been made.

5.2 Figure 3 and Figure 4: crisis asset trade and coexistence

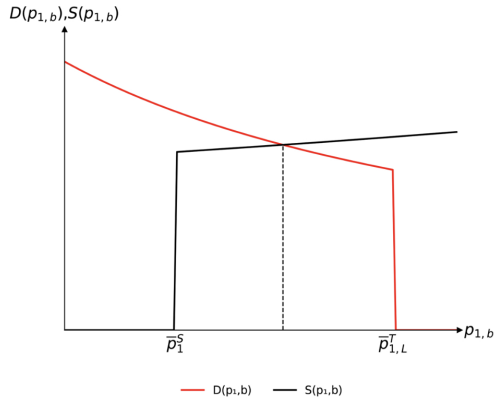
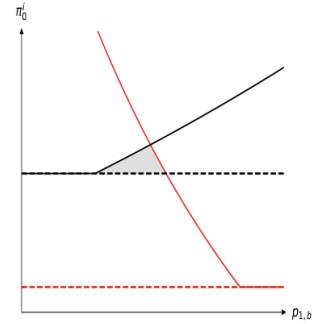


Figure 3

Numerical illustration of the asset market equilibrium in a crisis.

T-banks' demand in red is $D(p_{1,b}) = k/p_{1,b}(1-\chi^S)$ if $p_{1,b} < \bar{p}_{1,L}^T$ (Equation (10)), and S-banks' asset supply in black is $S(p_{1,b}) = n/(1-(1-\epsilon)p_{1,b})\chi^S$ if $p_{1,b} > \bar{p}_1^S$ (Equation (11)). The threshold values are $\bar{p}_{1,L}^T \equiv 1/(1 + \frac{p(\delta R - 1)}{(1-p)qR})$ and $\bar{p}_1^S \equiv 1/((1-\epsilon)(1 + \frac{p(R-1)}{(1-p)qR}))$. We use the following parameter values satisfying the assumptions made in Section 1.4: $n=1.5, p=0.78, q=0.99, r=1, R=2.3, k=2.7, \delta=0.65, \epsilon=0.1, \chi^S=0.86$.

A Case 1: $\pi_0^{T,NM} > \pi_0^{S,NM}$ absent asset trade



B Case 2: $\pi_0^{T,NM} < \pi_0^{S,NM}$ absent asset trade

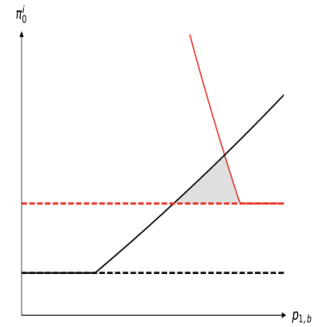


Figure 4

Numerical illustration of the coexistence of traditional and shadow banks. π_0^T is in red and π_0^S is in black. The plain lines correspond to T- and S-banks' profits when they trade assets in a crisis. The dashed lines correspond to T- and S-banks' profits when banks are not allowed to trade assets in a crisis. In that case, S-banks' expected profit is $\pi_0^{S,NM} = [pR + (1-p)qR]n$ and T-banks' expected profit is $\pi_0^{T,NM} = p[\delta R(n+k) - k] + (1-p)[\delta qR(n+k) - k]$, where we use the superscript "NM" for "no market". We use the following parameter values satisfying the assumptions made in Section 1.4: $n=1.5, p=0.78, q=0.99, r=1, R=2.3, \delta=0.65, \epsilon=0.1$. In Figure 4, panel A, we set $k=2.7$, whereas in Figure 4, panel B, $k=3.7$.

Read:

- Figure 3 plots T-bank asset demand and S-bank asset supply in the crisis state. The equilibrium crisis price is where the two curves intersect.
- Figure 4 compares bank expected profits with and without crisis asset trade. The shaded gap is the gain from trade that allows coexistence.
- The message is simple but important: without crisis asset trade, one bank type would dominate; with it, the two types can coexist.

5.3 Figures 5–7: liability flows out of shadow banks and into traditional banks

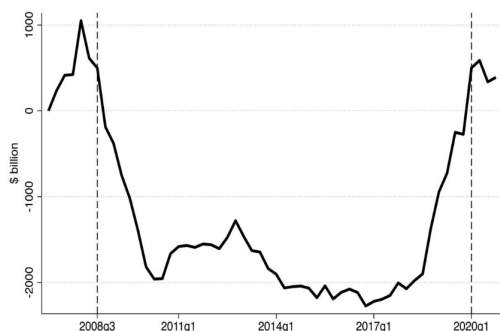


Figure 5
Shadow banks: Short-term debt cumulative flows since 2007q1.
We define short-term shadow bank debt as the sum of money market mutual fund shares (table L.206), federal funds and security repurchase agreements (table L.207), and open market paper net of traditional banks' holdings (table L.209). See [Internet Appendix B.1](#) for details. *Source:* Financial Accounts of the United States.

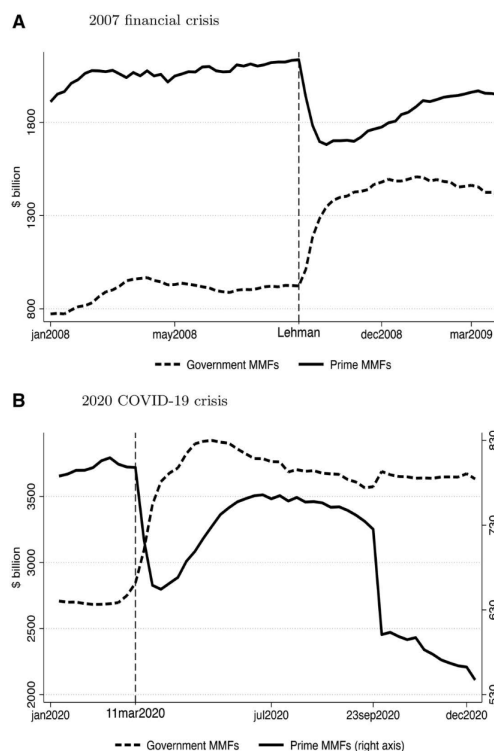


Figure 6
Shadow bank runs in 2007 and 2020.
The figure shows how during the 2007 and COVID-19 crises, investors ran from prime money market funds that lent to holders of MBS toward less risky government money market funds. A run on MMFs causes the funding markets for both commercial paper and bank certificates of deposit to be disrupted. *Source:* Investment Company Institute.

Read:

- Figure 5 shows a buildup and then collapse of shadow-bank short-term debt around both the 2007 and 2020 crises.
- Figure 6 shows runs from prime MMFs into safer government MMFs in both episodes, highlighting the fragility of short-term S-bank funding.
- Figure 7 shows very large deposit inflows into the largest U.S. traditional banks: about \$600 billion in the three weeks from September 10 to October 1, 2008, and more than \$800 billion in the three weeks after March 11, 2020.
- These figures show that crises involve a broad reallocation of safe-liability demand toward traditional banks.

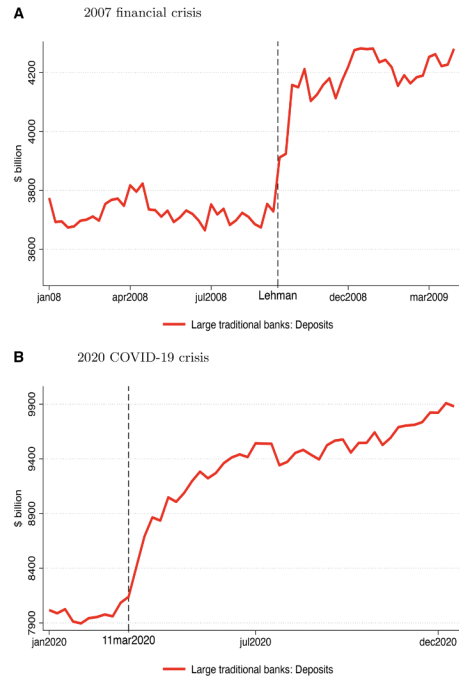


Figure 7
Inflow in deposits into the largest U.S. traditional banks.
We use the Federal Reserve's definition of the largest traditional banks: the top 25 domestically chartered commercial banks, ranked by domestic assets as of the previous commercial bank Call Report to which the H.8 release data have been benchmarked. *Source:* Fed H.8 Releases (weekly).

5.4 Figure 8 and Table 1: asset flow from shadow to traditional banks



Figure 8
Asset flow out of shadow banks and into traditional banks.
Agency- and CDO-backed securities holdings by traditional and shadow banks (% annual growth). We measure traditional banks' holdings using data on private depository institutions (L.110). Shadow banks' holdings are defined as the total holdings of those assets (L.211), excluding holdings by households, the government, traditional banks, and the Federal Reserve. See [Internet Appendix B.1](#) for details. *Source:* Financial Accounts of the United States.

Table 1
Traditional banks: Determinants of MBS purchases in the 2007 crisis.

	MBS purchases				
	Accounting for ABCP conduits consolidation				
	Scenario 1 (1)	Scenario 2 (2)	Scenario 1 (3)	Scenario 2 (4)	"Naive" scenario (5)
Change in uninsured deposits	0.01 (0.05)	0.01 (0.04)	0.05 (0.04)	0.05 (0.04)	0.06* (0.04)
Change in insured deposits	0.12*** (0.01)	0.12*** (0.04)	0.13*** (0.04)	0.13*** (0.04)	0.12*** (0.04)
Change in liquidity ratio	-0.16*** (0.04)	-0.17*** (0.04)	-0.16*** (0.04)	-0.16*** (0.04)	-0.17*** (0.04)
Change in credit	-0.11*** (0.04)	-0.10*** (0.05)	-0.10*** (0.05)	-0.10*** (0.05)	-0.10*** (0.05)
Capital ratio in 2007q4	-0.06 (0.05)	-0.05 (0.05)	-0.05 (0.05)	-0.05 (0.05)	-0.02 (0.05)
Unused commitments ratio in 2007q4	-0.00 (0.05)	-0.02 (0.05)	-0.01 (0.05)	-0.01 (0.05)	-0.07*** (0.05)
Controls	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	20	22	24	25	22
Observations	3,564	3,564	3,564	3,564	3,564

The unit of observation is the bank-holding company. We use White robust standard errors and winsorize all variables at the 1% level. Variable labels ending in 2007q4 represent variable levels as of 2007q4. Variable labels starting with "Change" are percentage growth rates in holdings from 2007q4 to 2009q4, as a function of 2007q4 total assets. The dependent variable MBS Purchases represents purchases of mortgage-backed securities by traditional banks between 2007q4 and 2009q4, normalized by total assets as of 2007q4. Similar to [Kling and Krishnamurthy \(2010\)](#), we approximate actual purchases of MBS by traditional banks from the change in their MBS holdings by testing a particular scenario for MBS repurchase rate and traditional bank asset losses (see [Internet Appendix B.2](#)). We report three of these scenarios. Under scenario 1 in column 1, we assume the repurchase rate (net of new issuances) is 7% and total losses imposed to the financial sector are \$500 billion. Under scenario 2 in column 2, the repurchase rate is 12% and total losses are \$176 billion. Finally, in the "naive" scenario in column 3, we use changes in end-of-year holdings without adjustment. Columns 4 through 6 use the same scenario while also accounting for ABCP conduits consolidation. To take ABCP conduits consolidation into account, we take the total amount of commercial paper issued by S-banks (see [Internet Appendix B.1](#)) and apportion it to individual banks according to their precise MBS holdings. Control variables are of three types. First, change variables: interest rates on large deposits, capital ratio, net wholesale funds, real estate, nonperforming loans ratio, maturity gap, interest rate on large deposits, net wholesale fund ratio, unused commitments ratio, real estate ratio, capital ratio. The second set of variables are stock variables as of 2007q4: liquidity ratio, nonperforming loans ratio, maturity gap, interest rate on large deposits, net wholesale fund ratio, unused commitments ratio, real estate ratio, capital ratio. The third type is a dummy variable for traditional banks' use of the "tag" (Transaction Account Guarantee) program, equal to one if the bank used the facility. *Source:* Call Reports and the [Fitch, Kling, and Krishnamurthy \(2010\)](#) columns. *p<0.1; **p<0.05; ***p<0.01.

Read:

- Figure 8 documents the core prediction of the model: assets move from S-banks to T-banks in crises.
- In 2020, S-banks reduced MBS holdings by roughly \$228 billion while T-banks increased theirs

by about \$110 billion. In the 2007 crisis, earlier work cited by the paper finds roughly \$800 billion of securitized-asset outflows from shadow banks and roughly \$550 billion of inflows into traditional banks.

- Table 1 shows that banks with bigger increases in *insured* deposits bought significantly more MBS during 2007–2009. The coefficient is roughly 0.22–0.23 and highly significant across specifications.
- This is the paper’s strongest model-to-data link: the banks that gained crisis-safe funding are the banks that absorbed shadow-bank assets.

5.5 Figures 9 and 10: fire sales and the Federal Reserve’s response

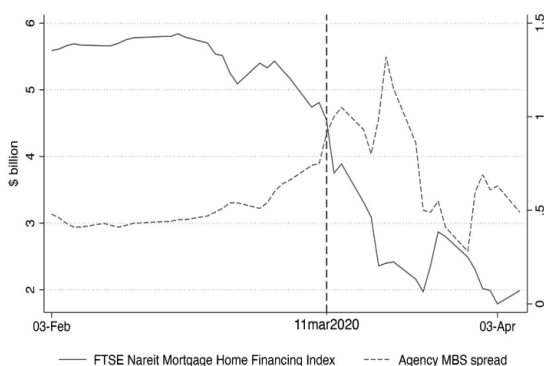


Figure 9
Asset fire sales during the 2020 COVID-19 crisis.
This figure illustrates the occurrence of fire sales during the COVID-19 crisis, when some shadow banks (mortgage real estate investment trusts) were forced to sell agency mortgage-backed securities (MBS) in order to satisfy increased margin calls. This selling pressure led to an increase in spreads on agency MBS (the dashed line plots the Bloomberg Barclays Agency Fixed-Rate MBS Average Option-Adjusted Spread) and a deterioration of shadow banks’ stock prices (the plain line plots the FTSE Nareit Mortgage Home Financing Index). This figure is inspired by [Sengupta and Xue \(2020\)](#). Source: Bloomberg.

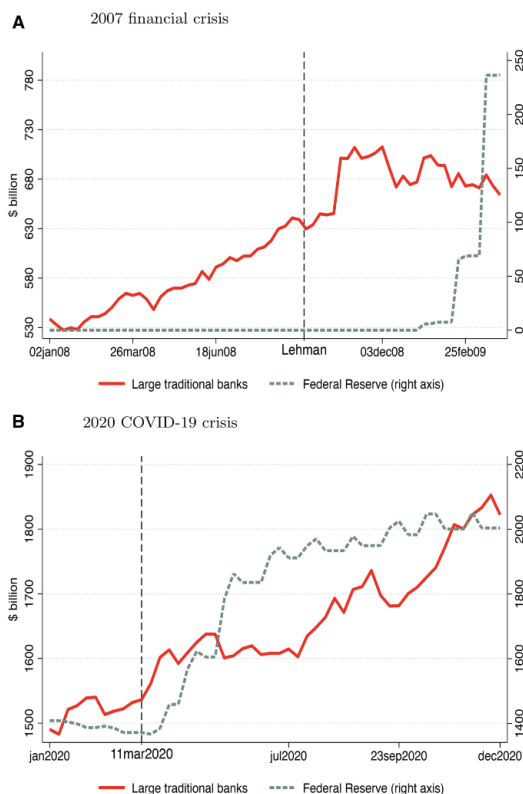


Figure 10
Large traditional banks’ MBS purchases and the Fed’s intervention.
This figure documents the timeline of the Fed’s purchase of agency MBS in response to the 2007 and COVID-19 crises. We show large traditional banks only due to the availability of weekly data for all other banks. We use the Federal Reserve’s definition of the largest traditional banks: the top 25 domestically chartered commercial banks, ranked by domestic assets as of the previous commercial bank Call Report to which the H.8 release data have been benchmarked. Source: Fed H8 Releases (weekly).

Read:

- Figure 9 shows fire-sale pressure in 2020: agency-MBS spreads rose sharply while mortgage-REIT valuations fell, consistent with forced selling by shadow intermediaries.
- Figure 10 shows that in 2008 the flow from S-banks to T-banks began before the Fed’s MBS intervention. The Fed only began purchasing agency MBS on November 25, 2008.
- In 2020 the Fed intervened much faster, but still after T-banks had already begun purchasing MBS. From March to December 2020, the Fed bought about \$500 billion of MBS while T-banks

bought about \$300 billion.

- The point is not that the Fed was irrelevant. It is that the private-sector T-bank/S-bank mechanism was already active and remains central to the coexistence story.

5.6 Figures 11–14: coexistence conditions, regulation, crisis likelihood, and welfare

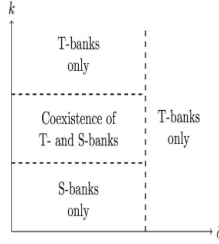
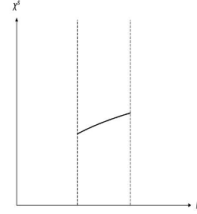


Figure 11
When do traditional and shadow banks coexist?
This figure illustrates how conditions (6) and (7) determine the possibility for T- and S-banks to be complementary and coexist. First, condition (6) ensures that the regulatory cost is high enough (δ low enough) for T-banks to be willing to keep slack at $t=0$ to purchase S-banks' assets in a crisis. Second, condition (7) ensures that the limit on T-banks' deposit insurance (k) is neither too low nor too large. Otherwise, T-banks' expected profit would always be larger or smaller than S-banks', respectively.

A Changes in T-banks' debt capacity



B Changes in T-banks' regulatory cost

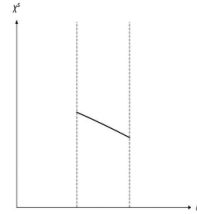


Figure 12
Numerical illustration of the (unintended) effects of regulations for traditional banks on the shadow banking sector.
We use the following parameter values satisfying the assumptions made in Section 1.4: $n=1.5, p=0.78, q=0.99, r=1, R=2.3, \varepsilon=0.1$. Figure 12, panel A, shows how the equilibrium fraction of S-banks (x^S) varies with $k \in [0, 6]$, and Figure 12, panel B, shows how this fraction varies with $\delta \in [0.76, 1]$. The dashed vertical lines represent the parameter set outside of which only one bank type exists in equilibrium.

Read:

- Figure 11 shows the coexistence region. If regulation is too light or T-bank debt capacity is too extreme, the model collapses to one bank type.
- Figure 12 shows the paper's key comparative statics: increasing T-bank debt capacity k can raise the equilibrium share of S-banks, while higher T-bank regulatory costs also raise S-bank prevalence.
- Figure 13 shows that S-bank prevalence is hump-shaped in the probability of the good state. If crises are too rare or too frequent, coexistence breaks down.
- Figure 14 shows that decentralized coexistence involves too many S-banks. Aggregate banker welfare peaks at a lower S-bank share than the market equilibrium chooses.

6 Short summary

- **Conclusion.** The paper explains the coexistence of traditional and shadow banks through crisis-period asset trade. T-banks provide the backstop that lets S-banks issue money-like claims even without deposit insurance.

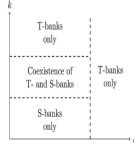


Figure 11
When do traditional and shadow banks coexist?
This figure illustrates how conditions (6) and (7) determine the possibility for T- and S-banks to be complementary and coexist. First, condition (6) ensures that the regulatory cost is high enough (if low enough) for T-banks to be willing to keep slack at $t=0$ to purchase S-banks' assets in a crisis. Second, condition (7) ensures that the limit on T-banks' deposit insurance (δ) is neither too low nor too large. Otherwise, T-banks' expected profit would always be larger or smaller than S-banks', respectively.

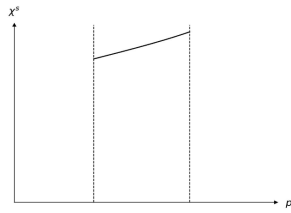


Figure 13
Numerical illustration of the effect of changes in the probability of a crisis on the share of shadow banks in equilibrium.
We use the following parameter values satisfying the assumptions made in Section 1.4: $n=1.5, q=0.99, r=1, R=2.3, k=2.7, \delta=0.65, \varepsilon=0.1$, and we vary $p \in [0.45, 1]$. The dashed vertical lines represent the parameter set outside of which only one bank type exists in equilibrium.

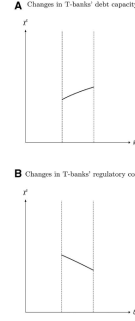


Figure 12
Numerical illustration of the (unintended) effects of regulations for traditional banks on the shadow banking sector.
We use the following parameter values satisfying the assumptions made in Section 1.4: $n=1.5, p=0.78, q=0.99, r=1, R=2.3, \varepsilon=0.1$. Figure 12, panel A, shows how the equilibrium fraction of S-banks (x^S) varies with $\delta \in [0.6]$, and Figure 12, panel B, shows how this fraction varies with $\delta \in [0.78, 1]$. The dashed vertical lines represent the parameter set outside of which only one bank type exists in equilibrium.

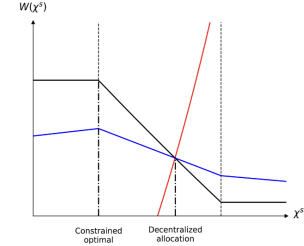


Figure 14
Numerical illustration of the socially excessive size of the shadow banking sector in the decentralized equilibrium.
 x^S_1 to x^S_2 in black, aggregate surplus in blue. We use the following parameter values satisfying the assumptions made in Section 1.4: $n=1.5, p=0.78, q=0.99, r=1, R=2.3, k=2.7, \delta=0.65, \varepsilon=0.1$, and we vary $x^S \in [0.81, 0.91]$. The red line represents x^S_1 ; the black line represents x^S_2 and the blue line represents the aggregate welfare of T- and S-banks as defined in (77). The dashed vertical lines represent the parameter set outside of which only one bank type exists in the decentralized equilibrium.

- **Identification insight.** The paper's evidence is built around matching a set of crisis facts to the model's unique mechanism: liability runs on S-banks, asset purchases by T-banks, and fire-sale pricing.
- **Main economic lesson.** Regulation of traditional banks affects shadow banking not only through substitution but also through complementarity. Stronger T-banks can make S-banks more viable.
- **Welfare lesson.** Market equilibrium overallocates bankers to the shadow sector because entry decisions do not internalize their effect on crisis asset prices.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the coexistence of traditional and shadow banks can be an equilibrium outcome of a simple model in which the two sectors interact in the crisis asset market.
- Deposit insurance does not eliminate shadow banking. Instead, it can create the conditions under which shadow banking becomes feasible, because insured banks stand ready to absorb risky assets in bad times.
- The same mechanism that supports coexistence also explains why crises feature simultaneous outflows of liabilities and assets from shadow banks into traditional banks.

7.2 Economic intuition

- Shadow banks are attractive in normal times because they avoid regulation.
- Traditional banks are attractive because they can issue safe claims in stress.
- The interaction between these two advantages creates complementarity. Shadow banks depend on traditional banks' balance sheets in crises, and traditional banks profit from shadow banks' need to sell.
- Once this crisis-market interaction is present, changing traditional-bank regulation affects not only traditional banks' own scale, but also the ex ante borrowing capacity of shadow banks.

7.3 Takeaway

This paper is best read as a coexistence theory of regulated and unregulated intermediation. Its main contribution is to show that shadow banking survives not in spite of traditional banking, but partly because traditional banks exist and can absorb fire-sold assets in crises. That same complementarity creates an important regulatory warning: policies that expand the crisis debt capacity of traditional banks can unintentionally support shadow banking rather than shrink it. The paper therefore gives a clear framework for thinking about deposit insurance, liquidity regulation, central-bank backstops, and the welfare costs of letting too much intermediation migrate into fragile shadow structures.